

Fig. 24.5. Cooper stares at the dust pattern on the floor of Murph's bedroom. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

In my science interpretation of *Interstellar*, I presume that Professor Brand's team has collected a large trove of data on the anomalies. The most interesting data to me as a physicist, and to Professor Brand in my movie interpretation, is new and changing patterns of *tidal gravity*.

We first met tidal gravity in Chapter 4: the tidal gravity produced by a black hole, and tidal gravity on Earth produced by the Moon and Sun. In Chapter 17 we saw Gargantua's tidal gravity in action on Miller's planet, triggering gigantic "Millerquakes," tsunamis, and tidal bores. In Chapter 16 we met the tiny stretching and squeezing of tidal gravity in a gravitational wave.

Tidal gravity is produced not only by black holes, the Sun, the Moon, and gravitational waves but also, in fact, by all gravitating objects. For example, regions of the Earth's crust that contain oil are